

RAIPUR, India

WMO: 428740

Lat: 21.233N

Lon: 81.650E

Elev: 298

StdP: 97.80

Time Zone: 5.50 (IND)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	12.7	13.8	1.8	4.5	27.9	3.5	5.0	27.1	3.4	22.4	2.9	23.1	0.3	50	0.312

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.6	43.2	22.5	42.1	22.7	40.9	22.5	27.2	32.1	26.7	31.4	26.5	31.0	2.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.1	22.3	29.3	25.7	21.8	28.7	25.4	21.4	28.4	88.4	32.4	86.5	31.4	85.1	31.2	32.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 5.4	4.4	3.6	DB	10.2	44.9	1.1	1.2	9.4	45.8	8.8	46.5	8.1	47.2	7.3	48.0	(4)
(5)			WB	7.9	28.4	1.2	0.8	7.1	28.9	6.4	29.4	5.8	29.8	4.9	30.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	27.5	21.4	24.5	28.7	32.7	35.1	31.8	27.8	27.4	27.9	26.8	24.0	21.4	(6)	
	DBStd	4.64	2.03	2.45	2.52	2.33	2.04	3.66	1.84	1.42	1.19	1.52	1.84	1.85	(7)	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD18.3	3	2	1	0	0	0	0	0	0	0	0	0	1	(9)	
	CDD10.0	6378	354	407	578	682	779	656	553	538	537	521	421	353	(10)	
	CDD18.3	3339	97	174	320	432	521	406	294	280	287	263	171	95	(11)	
	CDH23.3	41261	754	1699	4093	6738	8668	5879	3106	2776	3029	2511	1335	672	(12)	
	CDH26.7	21944	189	761	2316	4471	6210	3635	1059	781	1027	939	431	127	(13)	
(14) Wind	WSAvg	1.2	0.6	0.9	1.0	1.4	1.6	2.0	2.0	1.7	1.2	0.8	0.5	0.5	(14)	
(15) Precipitation	PrecAvg	1237	11	13	18	13	21	174	368	334	218	56	6	7	(15)	
	PrecMax	1646	109	72	47	111	113	633	698	520	444	175	60	67	(16)	
	PrecMin	794	0	0	2	0	0	23	110	162	41	2	0	0	(17)	
	PrecStd	230	24	17	11	22	25	122	143	96	96	50	13	16	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.8	35.6	39.6	43.0	45.0	44.1	36.1	33.2	33.8	33.8	32.3	30.6	(19)	
		MCWB	18.7	19.1	20.4	21.5	22.2	23.0	26.2	26.1	25.2	23.3	21.7	19.5	(20)	
	2%	DB	29.9	33.8	38.2	41.9	43.9	42.5	33.8	32.1	32.7	32.7	31.1	29.2	(21)	
		MCWB	18.5	19.4	20.6	21.7	22.1	23.7	26.0	26.0	25.5	23.3	21.2	18.8	(22)	
	5%	DB	28.7	32.5	37.0	40.9	42.8	40.8	32.4	31.2	31.9	31.9	30.2	28.1	(23)	
		MCWB	18.0	19.1	20.5	21.4	22.5	24.3	26.1	26.0	25.7	23.5	20.7	18.3	(24)	
	10%	DB	27.5	31.1	35.7	39.8	41.6	38.7	31.2	30.3	31.1	31.0	29.3	27.0	(25)	
		MCWB	17.4	18.9	20.3	21.4	22.7	24.6	26.0	25.8	25.7	23.2	20.3	17.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.1	23.4	24.7	26.0	26.9	28.0	28.1	27.2	27.4	26.5	24.6	21.9	(27)	
		MCDB	26.2	29.9	35.4	37.6	38.4	35.1	32.2	30.7	30.9	30.3	29.6	27.3	(28)	
	2%	WB	20.4	21.8	23.4	25.0	26.0	27.1	27.0	26.8	26.8	26.0	23.7	21.0	(29)	
		MCDB	25.6	29.8	33.6	36.3	37.4	33.8	31.2	30.1	30.5	29.9	28.0	26.1	(30)	
	5%	WB	19.4	20.8	22.5	24.1	25.3	26.6	26.6	26.5	26.5	25.5	22.9	20.1	(31)	
		MCDB	25.9	28.6	32.3	35.7	36.6	33.1	30.5	29.6	30.0	29.3	27.0	24.9	(32)	
	10%	WB	18.6	20.1	21.7	23.3	24.7	26.1	26.2	26.1	26.2	25.0	22.1	19.2	(33)	
		MCDB	25.6	27.9	31.6	34.8	36.0	32.6	29.9	29.1	29.6	28.6	26.7	24.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	12.4	12.8	13.2	12.6	9.0	5.3	4.8	6.1	8.6	11.5	12.3	(35)	
		MCDBR	13.7	13.9	14.2	14.1	13.5	11.8	7.5	6.3	7.4	9.7	11.9	13.3	(36)	
	5% WB	MCWBR	5.2	4.3	4.2	3.8	3.4	2.6	2.3	2.1	2.5	2.9	4.1	5.1	(37)	
		MCWBR	11.1	11.3	12.2	12.8	12.1	9.1	6.4	5.7	6.4	7.1	8.8	10.4	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.467	0.477	0.536	0.557	0.602	0.658	0.597	0.557	0.550	0.530	0.516	0.475	(40)	
		taud	2.016	2.010	1.888	1.887	1.813	1.737	1.880	2.004	2.003	1.994	1.942	2.010	(41)	
	Ebn at Noon		791	813	782	771	731	685	729	762	760	756	737	765	(42)	
		Edh at Noon	160	171	201	204	218	233	202	179	177	172	172	157	(43)	
(44) All-Sky Solar Radiation	RadAvg		4.53	5.35	6.18	6.83	6.85	5.23	3.97	3.92	4.68	5.04	4.63	4.30	(44)	
	RadStd		0.24	0.23	0.30	0.25	0.16	0.53	0.47	0.37	0.45	0.40	0.26	0.22	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page